The run image

"It works on my machine." "Then we will ship your machine."

A Docker image holds a small Linux system with a program and all that it needs, so it runs the same on any computer.

Stage	Holds
`uv`	uv, from its own image
`build`	the dependencies of the server, from `uv.lock`, in `/opt/venv`
`run`	Python, `/opt/venv` and the code. It starts uvicorn

The run image has no uv, no tests and no ruff. It is 173 MB.

The run image

"It works on my machine." "Then we will ship your machine."

A Docker image holds a small Linux system with a program and all that it needs, so it runs the same on any computer.

Stage	Holds
`uv`	uv, from its own image
`build`	the dependencies of the server, from `uv.lock`, in `/opt/venv`
`run`	Python, `/opt/venv` and the code. It starts uvicorn

The run image has no uv, no tests and no ruff. It is 173 MB.

System tests

Real HTTP requests go to the server in the run image, the server as it runs for the users of the app.

Recipe	Does	Needs a server
`just test`, `just check`	the unit and integration tests	no
`just docker-serve`	starts the run image at port 9000	
`just test-system`	sends the requests	yes, and fails without one
`just docker-stop`	stops the server	

A server is a fixture of the commands that need it. Start it, use it, and stop it.

Two images from one lockfile

Image	Holds	Starts	Size
run	the server and its dependencies	uvicorn	173 MB
dev	the project, every group, uv, just and git	a shell	353 MB

`just docker-test` runs `just check` in the dev image, with this folder mounted at `/work`.

A new endpoint

`/add/` adds two numbers, and an integration test calls it.

1. `just check` on this machine
2. `just docker-test` in the dev image
3. `just docker-build run` builds the run image again, with the new code, and `just docker-serve` starts it
4. `just test-system` against the new run image, then `just docker-stop`